Unit 2 Guided Notes — Scientific Notation

Big or tiny number? Pack it into a × 10^n.

Standards: NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

1 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \le a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45{,}000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent −3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	_____	$45{,}000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	_____	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE A Scientific Notation Conversions

Remember: 1 ≤ a < 10. Positive exp = BIG · Negative exp = SMALL.

1

45,000,000 → sci. notation

2

0.00032 → sci. notation

3

6.02×10^8 → standard

4

9.1×10^{-5} → standard

5
780,000 → sci. notation

6
0.0000067 → sci. notation

7
3.45×10^{4} → standard

8
1.2×10^{-3} → standard

9
5,280,000,000 → sci. notation

10
2.5×10^{-2} → standard

11
0.000000904 → sci. notation

12
7.15×10^{6} → standard

13
63,400,000,000 → sci. notation

14
8.8×10^{-6} → standard

Multiplying Scientific Notation — 3-Step Process

1. _____ the coefficients (the a-values)
2. _____ the exponents
3. Adjust if coefficient is NOT between _____ and _____

Example (no adjustment): $(3 \times 10^{4}) \times (2 \times 10^{5})$

Multiply coefficients: 3 × 2 = 6

ADD exponents: 4 + 5 = 9

✓ **Answer:** 6×10^{9}

Dividing Scientific Notation — 3-Step Process

1. _____ the coefficients
2. _____ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ **ADJUSTMENT RULES:**
- If coefficient ≥ 10: Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1: Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^5

Adjust (20 ≥ 10): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3. Small number → NEGATIVE exponent. Correct: **4.5×10^{-3}**. Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE B Operations with Scientific Notation

Multiply/divide coefficients · add/subtract exponents · adjust so $1 \leq a < 10$.

1
$(3 \times 10^4)(2 \times 10^5)$

2
$(8 \times 10^7) \div (4 \times 10^3)$

3
$(5 \times 10^3)(7 \times 10^6)$

4
$(9.6 \times 10^8) \div (3.2 \times 10^5)$

5
$(6 \times 10^5)(5 \times 10^4)$

6
$(1.5 \times 10^{-3})(4 \times 10^7)$

7
$(7.2 \times 10^9) \div (1.8 \times 10^{-2})$

8
$(2.5 \times 10^6)(8 \times 10^{-3})$

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE C Word Problems

Identify the operation, write your setup, then adjust so $1 \leq a < 10$.

1. The Sun is about 9.3×10^{7} miles from Earth. Light travels 1.86×10^{5} miles per second. How many seconds for light to reach Earth?

Answer: _____

2. A bacterium is about 2×10^{-6} m long; a virus about 5×10^{-8} m. How many times longer is the bacterium?

Answer: _____

3. The US debt is about 3.4×10^{13} dollars; population about 3.4×10^{8}. What is the debt per person?

Answer: _____

4. A grain of sand has mass 6.5×10^{-4} g. A beach has 8×10^{18} grains. Total mass in grams?

Answer: _____

5. A body has 3.7×10^{13} cells, each 1×10^{-5} m wide. Lined up, how long in meters?

Answer: _____

6. A computer does 4.8×10^{9} calculations per second. A task needs 1.44×10^{12}. How many seconds?

Answer: _____

7. A hard drive holds 2×10^{12} bytes. Each photo is 4×10^{6} bytes. How many photos fit?

Answer: _____

8. Light travels 3×10^{8} m/s. How far in 6×10^{3} seconds?

Answer: _____

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Scientific Notation Conversions

1
7,200,000 → sci. notation

2
0.00045 → sci. notation

3
3.05×10^{7} → standard

4
6.1×10^{-5} → standard

5
91,000 → sci. notation

6
0.0000008 → sci. notation

7
4.44×10^{6} → standard

8
2.7×10^{-3} → standard

REVIEW · B Operations

1
$(3 \times 10^{4})(3 \times 10^{5})$

2
$(8 \times 10^{9}) \div (2 \times 10^{3})$

3
$(5 \times 10^{6})(4 \times 10^{3})$

4
$(9.6 \times 10^{8}) \div (1.6 \times 10^{2})$

5
$(7 \times 10^{5})(8 \times 10^{2})$

6
$(1.2 \times 10^{-3})(4 \times 10^{5})$

REVIEW · C

Word Problems

1. A red blood cell is about 8×10^{-6} m wide; a platelet about 2×10^{-6} m. How many times wider is the red blood cell?

Answer: _____

2. Earth is about 1.5×10^{8} km from the Sun. A spacecraft travels 3×10^{4} km/h. How many hours for the trip?

Answer: _____

3. A factory makes 5×10^{3} widgets per day. How many in 2×10^{2} days?

Answer: _____

4. A virus is about 3×10^{-7} m and a cell about 6×10^{-5} m. How many times larger is the cell?

Answer: _____